Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/15/2008
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Project Title:	Impaired Activation and Signal Transduction in Mouse Splenocytes Cultured in Modeled Microgravity		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
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Task Description:	Astronauts experience decreased cellular immunity during spaceflight. To successfully achieve long-term manned spaceflight, countermeasures must be developed to ameliorate immune function during these missions. The design of effective countermeasures will require a detailed knowledge of the underlying molecular mechanisms responsible for decreased immunity in spaceflight. T lymphocytes, the principle mediators of cellular immunity, are insensitive to antigen-receptor stimulation when cultured in vitro during orbital flight. The inhibited activation of T cells during spaceflight is microgravity dependent and the cells remain partially responsive to direct activation of the diacyl glycerol (DAG) and calcium signaling pathways that lie downstream of the T cell antigen-receptor (T cell receptor, TCR). A similar pattern of inhibition is observed when T cells are cultured in ground-based microgravity analogs such as rotating-wall vessel (RWV) bioreactors. Studies in the RWV have specifically implicated alterations in DAG signaling as being responsible for the inhibited activation of T cells in microgravity. This project investigates the hypothesis that microgravity prevents T cell activation by hampering the DAG signal that is generated in response to stimulation of the		

	TCR. Specifically we have cultured primary human T cells in the RWV to investigate the effects of microgravity on the TCR-induced generation of the DAG signal and also the activation of DAG-responsive signaling elements downstream of the TCR. The ultimate goal of these investigations is to identify the underlying molecular mechanisms responsible for the inhibitory effect of MMG on T lymphocyte activation for the purpose of evidence-based design of countermeasures to ameliorate the immune response during long-term spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In vitro culture of T cells in true- and modeled-microgravity provides a unique (non-chemical, epigenetic) means of inhibiting signal transduction; allowing for a minimally invasive examination of the key signaling mechanisms required for T cell activation. Study of these mechanisms in fully functional cells in the severely inhibitory environment of the RWV will also provide insight into T cell function in less inhibitory situations such as aging. Conversely, identification of key signaling mechanisms in these pathways could also provide novel targets for treatment of autoimmune disease.
Task Progress:	The working hypothesis of this project is that microgravity prevents T cell activation by disrupting the transmission of activating signals from the T cell receptor (TCR) to the interior of the cell. To investigate this hypothesis we proposed (1) to investigate the temporal aspects of inhibition, (2) to determine whether or not microgravity differentially affects the T cell subsets, and (3) to directly measure TCR signaling in the RWV with the goal of locating a specific point in the antigen-receptor signal that is disrupted by microgravity. The research conducted over the last year completes the second and third of these aims, and brings this project to its conclusion. In concluding this research the student-recipient of this GSRP was awarded his PhD, and we are currently preparing two publications for submission to peer reviewed journals early in 2008. We began our investigation of the differential effects of microgravity on T cell subsets during the last reporting period, and found that the helper (Th) and cytotoxic (Tc) T cell subsets were equally inhibited by culture in the RWV. During this reporting period we focused our investigations on measuring differences between naïve and memory Th cells. This was accomplished by taking advantage of the ability of phorbol esters to rescue the TCR-induced activation of a subset of T cells in the RWV. In these experiments we stimulated primary human peripheral blood with a T cell-specific mitogenic lectin in the presence or absence of a sub-mitogenic dose of phorbol ester (PMA), and compared the subsequent activation of T cells in 1-g and RWV culture. In these experiments we found that although PMA co-stimulation activates a small fraction of both naïve and memory Th cells, cells of the latter subset are activated more effectively. We conclude that the preferential rescue of activation in the memory subset reflects the lowered activation threshold of this subset of cells compared to their naïve counterparts. In the final aim of this project we proposed to directly measure a defect in the T cell antigen-receptor signal during culture in microgravity. Based on previous observations by ourselves and others we chose to specifically focus on signaling to and through the diacyl glycerol (DAG) second-messenger pathway in these investigations. DAG is generated by phospholipase Cg1 (PLCg1), and in turn induces signaling through the mitogen activated protein kinase (MAPK) and nuclear factor kappa B (NFkB) pathways. We therefore began our investigations by comparing the ability of T cells cultured in 1-g or RWV conditions to transduce activating signals from the T cell receptor (TCR) to PLCg1. This was accomplished using flow cytometry and Western blots to measure the tyrosine phosphorylation of several key signaling molecules whose activation is required for the TCR-induced activation of PLCg1. Somewhat surprisingly we found that culture in the RWV did not prevent phosphorylation of any of the signaling molecules we examined, and importantly, PLCg1 was activated to the same extent in the RWV as in 1-g conditions. These results indicate that antigen-receptor stimulation of T cells is as effective in the RWV as at 1-g, and that microgravity must impinge upon the activation pathway at a point downstream of PLCg1 and generation of the DAG second-messenger. DAG induces signaling through NFkB by activating protein kinase Ctheta (PKCtheta). We began our investigation of signaling through this pathway by confirming that PKCtheta was activated by T cells stimulated in the RWV. PKCtheta has a diverse role in T cell activation, and in addition to inducing NFkB signaling this enzyme participates in the cytoskeletal rearrangements and TCR-modulation that accompany activation. When these two aspects of T cell activation were used as surrogate markers for PCKtheta activity we found that T cells cultured in the RWV were not inhibited in either their ability to polymerize the actin cytoskeleton, or to downregulate surface-expression of the TCR in response receptor ligation. These results clearly indicate that PKCtheta is activated in the RWV; however, whether NFkB signaling is effectively induced by this activity is less clear. When we used Western blots to directly measure signaling to NFkB we found that this signal was induced, but possibly with delayed kinetics and magnitude. DAG induces signaling through the MAPK pathway by activating the guanyl-nucleotide exchange factor for RAS (RASGRP). Activation of RAS by RASGRP initiates a MAPK signaling cascade that induces the enzymatic activity of the extracellular regulated kinase (ERK). ERK is the primary effector of this signaling pathway, and we first determined if this enzyme was activated in response to TCR-ligation during culture in the RWV. ERK enzymatic activity only requires phosphorylation at two residues, and we found dually phosphorylated ERK to be equally enriched in lysates of T cells stimulated in RWV and 1-g conditions. ERK is a promiscuous enzyme, and a critical aspect of its regulation is access to substrates. Therefore we next sought to determine if culture in the RWV altered the sub-cellular distribution of activated ERK. The data generated in these experiments indicates that in the RWV ERK may become sequestered in the cytosol, and thus be prevented from activating nuclear substrates. Our data thus indicate that the initiation of the TCR signal and generation of the DAG second-messenger are not affected by culture in the RWV. Transmission of this signal to the nucleus, however, appears to be inhibited by microgravity. While our data indicate a possible role for delayed NFkB signaling in preventing T cell activation in the RWV, a dominant role for altered MAPK signaling is more strongly implicated. We conclude that defective T cell activation in the RWV is due to the absence of ERK-induced transcriptional activity in the nucleus.
Bibliography Type:	Description: (Last Updated: 01/15/2008)
Dissertations and Theses	Simons DM. "A Dissection of T cell Receptor Signaling Pathways in Primary Human T cells Activated in the Rotating-Wall Vessel Bioreactor." Dissertation, Drexel University, Philadelphia, PA, December 2007. , Dec-2007